Discrete - Time LTI Systems

Scaled, Delayed DT unit impulse $\alpha\delta[n-k]$

As learned previously, $\delta[n]$ is a kick of height 1 at $n=0$.
However with some calculus, we can easily generalise it more.

$$\delta[n] = \begin{cases} 1 & \text{if } n=0 \\ 0 & \text{if } n\neq 0 \end{cases} \xRightarrow{\text{Generalised}} \alpha\delta[n-k] = \begin{cases} \alpha & \text{if } n=k \\ 0 & \text{if } n\neq k \end{cases}$$

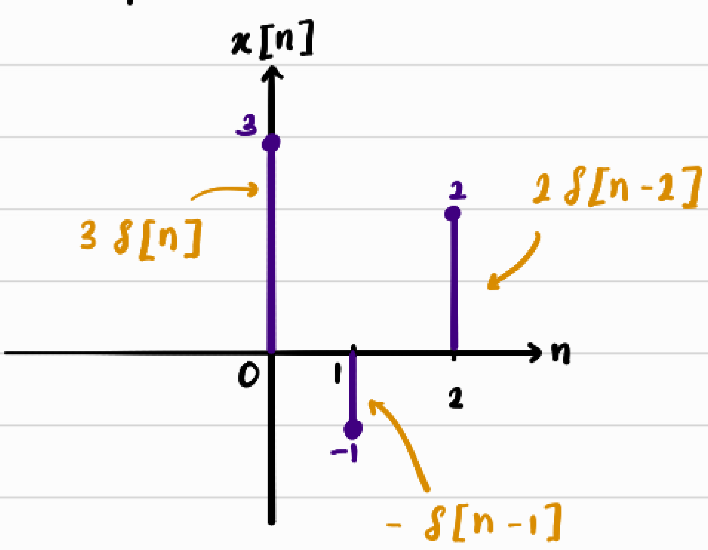
$\alpha\delta[n-k]$ just means put value/sample of height α at k

Discrete - Time Sifting formula

Since every DT signal is just a collection of samples, we can represent each samples with scaled (weighted), delayed (shifted) impulse.

$$\boxed{x[n] = \sum_{k=-\infty}^{\infty} \overbrace{x[k]}^{\text{Value to place}}\ \underbrace{\delta[n-k]}_{\text{position of value, } x[k]}}$$

Example



Concept :

n : Which signal do I want to reconstruct ?

k : Run through all samples to find the one that contributes

∴ Summing up the samples gives
$x[n] = 3\delta[n] - \delta[n-1] + 2\delta[n-2]$

We have just learned how to reconstruct signals, $x[n]$ using impulses. Now we run this signal into a system.

Impulse Response, $h[n]$

The impulse response, $h[n]$ is the system output when the input signal, $x[n]$ is the unit impulse, $\delta[n]$

Suppose

$$x \longrightarrow \boxed{\text{System}} \longrightarrow y$$

If we let $x[n] = \delta[n]$, then by definition $y[n]$ is the impulse response, $h[n]$.

$$\delta[n] \longrightarrow \boxed{\text{System}} \longrightarrow h[n]$$

> $h[n]$ is essentially telling us the system's complete response pattern to one unit input sample at $n = 0$

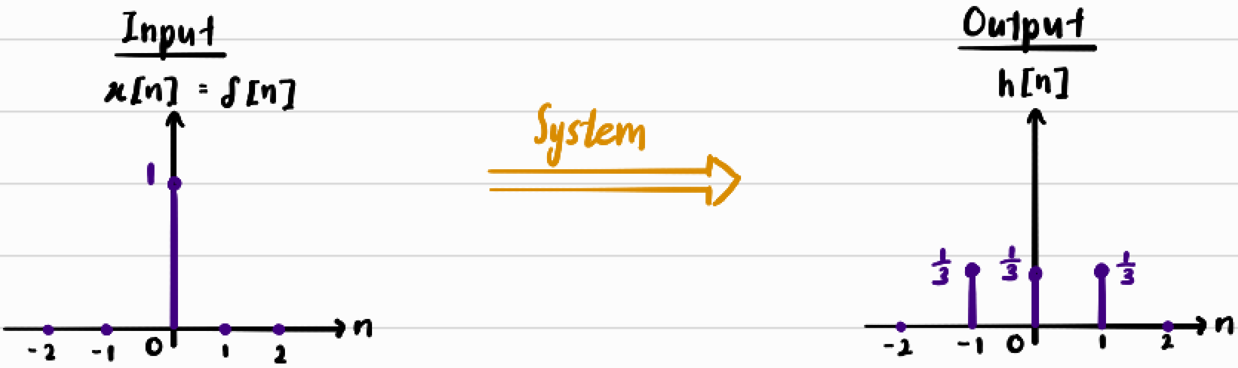
Example 1 [Non Recursive]

Find impulse responce, h to the 3-point avg system below

$$y[n] = \frac{1}{3}\left[x[n] + x[n-1] + x[n-2]\right]$$

let $x[n] = \delta[n]$

$$\Longrightarrow h[n] = \frac{1}{3}\left[\delta[n] + \delta[n-1] + \delta[n-2]\right]$$



> So $h[n]$ tells us that the system takes one sample and spread its effect across 3 output samples

Example 2

Find the impulse response, h of $y[n] = 0.5\, y[n-1] + x[n]$ with initial rest — $y[n] = 0,\ n < 0$

Let $x[n] = \delta[n] \implies h[n] = 0.5\, h[n-1] + \delta[n]$

At $n = 0$:

$$h[0] = 0.5\, h[-1] + \delta[0]$$
$$= 0 + 1$$
$$h[0] = 1$$

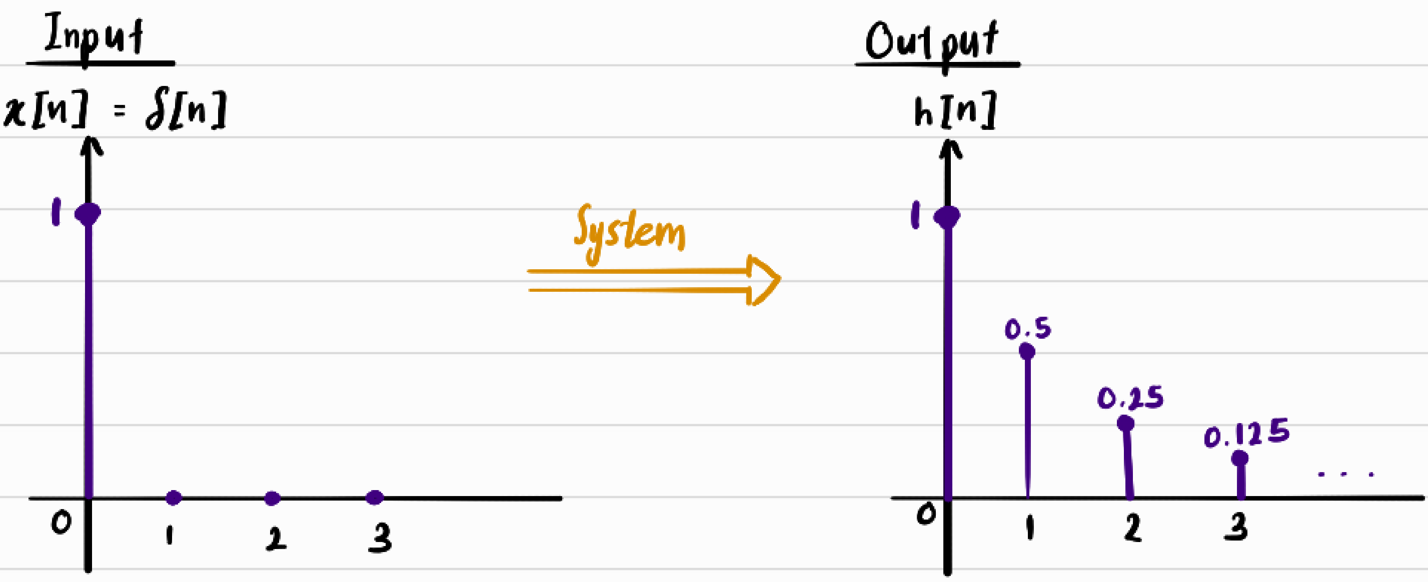
At $n = 1$:

$$h[1] = 0.5\, h[0] + \delta[1]$$
$$= 0.5\,[1] + 0$$
$$h[1] = 0.5$$

At $n = 2$:

$$h[2] = 0.5\, h[1] + \delta[2]$$
$$= 0.5\,[0.5] + 0$$
$$h[2] = 0.25$$

$$\therefore\ h[n] = (0.5)^n u[n]$$



The impulse only occured once, but cuz the system uses its previous output as an input, that one impulse input caused a "domino effect"

Some Extra Terminology

Finite Impulse Response (FIR)

$h[n]$ is non-zero finitely many times

E.g. Example 1 : 3-point avg $[n=1, 2, 3]$

Infinite Impulse Response (IIR)

$h[n]$ is non-zero infinately many times

E.g. Example 2 : $h[n] = (0.5)^n u[n]$

This begs the question : If I already know the system's response to $\delta[n]$, can I predict its response to every one of those shifted impulses, $\delta[n-k]$?

~ For an LTI system, Yes.

Step 1 : Shift / Delay the impulse

Because the system is time-invariant, waiting k samples before applying the input cannot change the system behaviour.
So its entire response also just delays by k samples

$$\boxed{\delta[n-k] \longrightarrow h[n-k]} \quad \text{——} \quad (1)$$

Step 2 : Scale the input impulse

Cuz the real input samples don't all have height 1, we scale the impulse by the actual sample height.

$$\boxed{\delta[n-k] \rightarrow x[k]\,\delta[n-k]}$$

Because of Linearity $[\, f(\alpha x) = \alpha f(x) \,]$

$$\boxed{x[k]\,\delta[n-k] \longrightarrow x[k]\,h[n-k]} \text{, using } (1)$$

Step 3 : Add every impulse together

Remember $x[n] = \sum_k x[k]\,\delta[n-k]$, so an arbitrary input contain lots of individual samples.

$$x[0]\,\delta[n] + x[1]\,\delta[n-1] + x[2]\,\delta[n-2] + \dots$$

And for each individual sample we do step 2

$$x[k]\,\delta[n-k] \longrightarrow x[k]\,h[n-k]$$

Because the system is linear, superposition holds so we add all those individual impulse response to each individual sample

Impulse response at k

Convolution
$[y[n] = (x*h)[n]]$

$$\boxed{y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]}$$

[System behavior at k of unit input impulse]

x convolved with h

Scales the system behaviour to the actual input sample value

Convolution Concept

Every input impulse, $\delta[n-k]$ creates its own copy of the impulse response where then the copy is...

1) Shifted to where the input sample occured
2) Scaled by the input sample's height

Then the output = sum of all those response copies.

We are essentially finding the pattern/behaviour ($h[n]$) from a single impulse at $n=0$, $\delta[n]$. Then cuz the system is LTI, we can freely scale and delay $h[n]$ in accordance to the "impulse interpretation" of input signal, $x[n]$

Example

Using the same 3-point avg eqn, we already know

$$h[n] = \frac{1}{3}[\delta[n] + \delta[n-1] + \delta[n-2]]$$

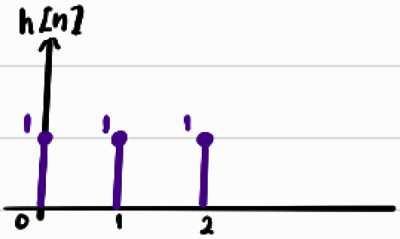
Let's assume we have this input signal

$$x[n] = 3\delta[n] - \delta[n-1] + 2\delta[n-2]$$

1^{st} sample : $3\delta[n]$

Since $\delta[n] \xrightarrow{\text{creates}} h[n]$, and our system is linear

$\Longrightarrow 3\delta[n] \xrightarrow{\text{creates}} 3h[n]$



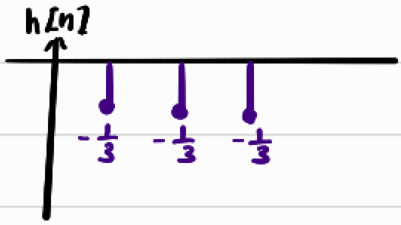
$$3h[n] = \delta[n] + \delta[n-1] + \delta[n-2]$$

2^{nd} sample : $-\delta[n-1]$

Since $\delta[n] \rightarrow h[n]$

$\delta[n-1] \rightarrow h[n-1]$ — Time Invariant

$-\delta[n-1] \rightarrow -h[n-1]$ — Linear



$$-h[n-1] = -\frac{1}{3}(\delta[n-1] + \delta[n-2] + \delta[n-3])$$

3^{rd} sample : $2\delta[n-2]$

Since $\delta[n] \rightarrow h[n]$

$\delta[n-2] \rightarrow h[n-2]$ — Time Invariant

$2\delta[n-2] \rightarrow 2h[n-2]$ — Linear

So finally to get $y[n]$, we just add up each samples's contribution.

$$\boxed{y[n] = 3h[n] - h[n-1] + 2h[n-2]}$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Graphical Interpretation of Convolution

This subtopic is just about convoluting graphically $x[n]$ and $h[n]$.

Theoractically, the transformation that take place are the following...

For a chosen n, we are given $x[k]$ and $h[k]$

$$h[k] \xrightarrow{\text{Reflect on } k=0} h[-k] \xrightarrow{\text{Shift right by } n} h[n-k] \xrightarrow{\text{Scale vertically by } x[k]} x[k]\,h[n-k] \xrightarrow{\text{Sum}} \boxed{y[n] = (x*h)[n] = \sum_k x[k]\,h[n-k]}$$

Take note

$h[-k] \rightarrow \underset{}{h[n-k]}$ is NOT shift left by n.

$h[-(k-n)]$

let $h[-k] = g(k)$

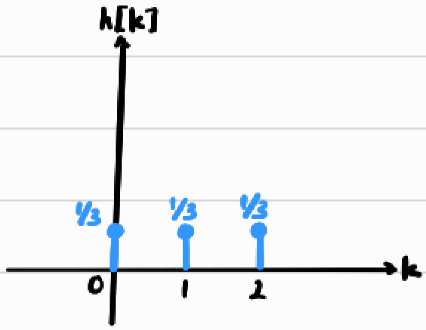
$h[-(k-n)] = g(k-n)$

Right shift by n

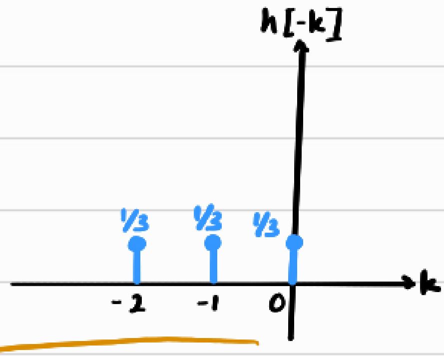
Example

Using back the 3-point avg, we have the following

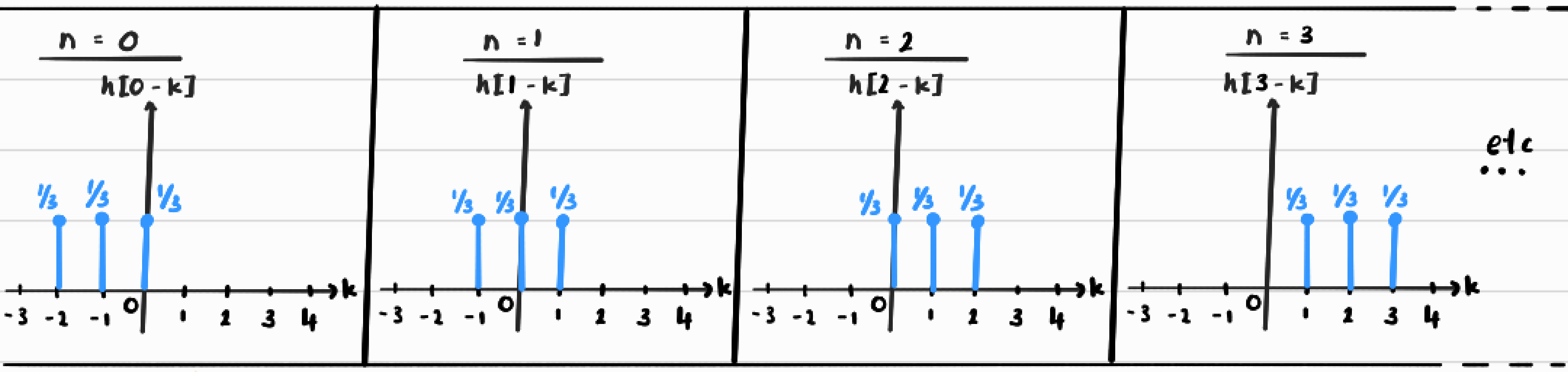
$$x[k] = 3\delta[k] - \delta[k-1] + 2\delta[k-2] \; ; \; h[k] = \tfrac{1}{3}(\delta[k] + \delta[k-1] + \delta[k-2])$$



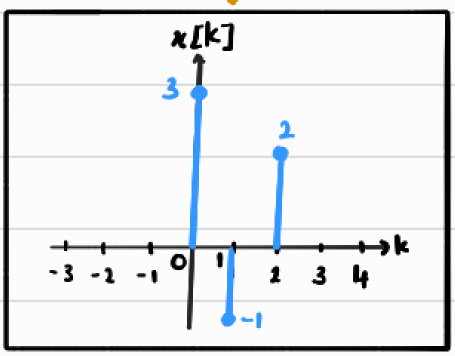
Flip about $k=0$
$h[-k]$

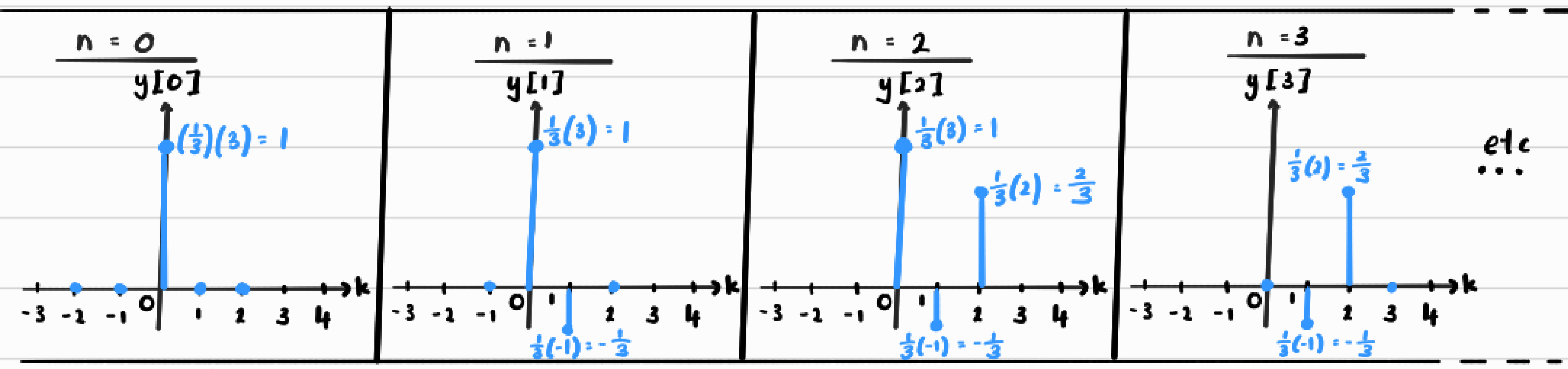


Shift right by n



Scale vertically by $x[k]$





Sum [Nxt Pg]

Table Form

	$x[0]$ $k=0$	$x[1]$ $k=1$	$x[2]$ $k=2$	Sum
$n=0$	1	0	0	$y[0] = 1$
$n=1$	1	$-1/3$	0	$y[1] = 1 + (-\frac{1}{3}) = \frac{2}{3}$
$n=2$	1	$-1/3$	$2/3$	$y[2] = 1 + (-\frac{1}{3}) + \frac{2}{3} = \frac{4}{3}$
$n=3$	0	$-1/3$	$2/3$	$y[3] = -\frac{1}{3} + \frac{2}{3} = \frac{1}{3}$
$n=4$	0	0	$2/3$	$y[4] = \frac{2}{3}$
$n=5$	0	0	0	$y[5] = 0$
$\vdots$		$\vdots$		$\vdots$

$$\therefore\ y[n] = \delta[n] + \frac{2}{3}\delta[n-1] + \frac{4}{3}\delta[n-2] + \frac{1}{3}\delta[n-3] + \frac{2}{3}\delta[n-4] \quad \#$$

Overall Concept Check :

We first "test the waters" of an LTI system with **one unit impulse** $\delta[n]$ to get the **system's basic response** $h[n]$ **/behaviour pattern**.

Because the system is Linear & Time Invariant, every **actual input** $x[k]$ **sample** produces its own **scaled & shifted copy of this system's basic response/behaviour pattern** $x[k]\,h[n-k]$.

At any particular time n, we collect the contribution from all **input samples** $x[k]$ and sum them up.

$$\boxed{y[n] = \sum_k x[k]\,h[n-k]}$$

k : Which input sample is contributing/overlapping

n : Which output sample we are finding

Sequence Table Method

still using the same 3-point avg where

$x[n] = 3\delta[n] - \delta[n-1] + 2\delta[n-2]$

$h[n] = \frac{1}{3}(\delta[n] + \delta[n-1] + \delta[n-2])$

	k	-2	-1	0	1	2	3	4
	$x[k]$	0	0	3	-1	2	0	0
	$h[k]$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	0
$n=0$	$h[-k]$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	0	0	0
$n=1$	$h[1-k]$	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	0	0
$n=2$	$h[2-k]$	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0	0
$n=3$	$h[3-k]$	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0
$n=4$	$h[4-k]$	0	0	0	0	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$

$y[0]$ $y[1]$ $y[2]$ $y[3]$ $y[4]$

$y[0] = x[k]\,h[-k]$
$= 3\left(\frac{1}{3}\right)$
$y[0] = 1$ #

$y[1] = x[k]\,h[1-k]$
$= 3\left(\frac{1}{3}\right) + -1\left(\frac{1}{3}\right)$
$= 2/3$

$y[2] = x[k]\,h[2-k]$
$= 3\left(\frac{1}{3}\right) + -1\left(\frac{1}{3}\right) + 2\left(\frac{1}{3}\right)$
$= 4/3$ #

$y[3] = x[k]\,h[3-k]$
$= -1\left(\frac{1}{3}\right) + 2\left(\frac{1}{3}\right)$
$= \frac{1}{3}$

$y[4] = x[k]\,h[4-k]$
$= 2\left(\frac{1}{3}\right)$
$= \frac{2}{3}$ #

Properties of Convolution

identity element

① $x * \delta = x$

Obviously, since you are just scaling each sample by 1.

② $x * h = h * x$ [Commutative]

It makes sense since it doesn't really matter who is sliding/shifting

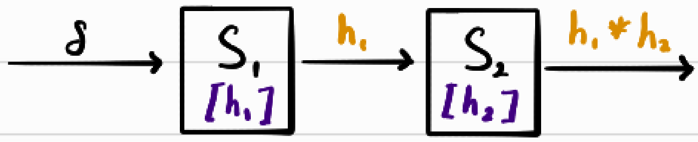
③ $h_1 * (h_2 * h_3) = (h_1 * h_2) * h_3$ [Associative]

④ $h_1 * (h_2 + h_3) = (h_1 * h_2) + (h_1 * h_3)$ [Distributive]

Convolution essentially is just sliding one signal over the other, multiplying then adding the results.
Since who sliding over who doesn't matter, convolution just obeys the properties of multiplication

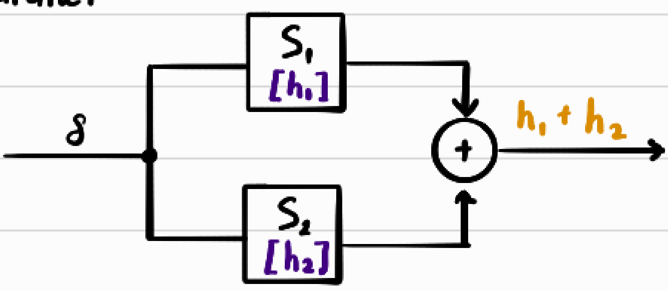
Series & Parallel LTI systems

① Series



$\boxed{h_{series} = h_1 * h_2}$

② Parallel



$\boxed{h_{parallel} = h_1 + h_2}$

Systems in general : $S_1(S_2(x)) \neq S_2(S_1(x))$
But for single input single output Discrete LTI systems,

$$h_1 * h_2 = h_2 * h_1 \quad \text{is true}$$

$$\therefore \quad S_1(S_2(x)) = S_2(S_1(x))$$

The impulse response is commutative, so is the systems.

Extracting system properties from impulse response, h[n]

Since h[n] completely decribes an LTI system, properties of the system must appear somewhere in h[n]

version 1

$$\boxed{y[n] = x * h = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]}$$

OR

version 2

$$\boxed{y[n] = h * x = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]}$$

Rmb that convolution is commutative

① **Memoryless** y[n] only depends on the current input, x[n]

Using version 2, for y[n] to only depend on x[n] $\Longrightarrow k = 0$

$\therefore$ That means only x[n] will not get scaled [and every other term = 0]

$$\Longrightarrow \boxed{h[k] = 0 \text{ for } k \neq 0}$$

Rewritten with impulse

$$\boxed{h[n] = \alpha\, \delta[n]}$$

$\therefore$ Every memoryless DT LTI system is just a gain system

$$y[n] = (\alpha\, \delta[n])(x[n])$$

$$y[n] = \alpha\, x[n]$$

② Causality

$y[n]$ only depends on the present or past

Using version 2, for $y[n]$ to not depend on the future $\Longrightarrow k \geq 0$

$\therefore$ That means all future signals will not get scaled

$\Longrightarrow$ $\boxed{h[n] = 0 \text{ for } n < 0}$

③ BIBO stability

Refering to version 2

Assuming the input signal is bounded, a system is BIBO stable if the impulse response (System behavior), $h[n]$ ...

1) Has finite valued samples

   However for our course, we are only working with ordinary finite-valued signal samples.

2) Has finitely many non-zero values. [Absolutely summable]

$$\boxed{\sum_{k=-\infty}^{\infty} |h[k]| < \infty}$$

Bcuz an infinite sum of finite values is still not bounded.

Take note:

As we learned previously, Finite Impuls Response (FIR) by definition has only finitely many non-zero values.

$\therefore$ Every FIR DT LTI system is BIBO stable

Continuous-Time LTI systems

Conceptually, CT LTI systems convolution works the same way, just that rather than summing (Σ) individual samples, we integrate ($\int$) continuously over time

Overall Translation

Discrete Time	Continuous time
n	t
dummy index, k	dummy index, τ
$\delta[n]$	$\delta(t)$
$\sum_k$	$\int d\tau$
$x[k]$	$x(\tau)$
$h[n-k]$	$h[t-\tau]$

For the following explanations, I'll only touch on things that DT and CT differs by.

Scaled, Shifted CT unit impulse

$$\delta(t) = \begin{cases} \infty & \text{if } t = 0 \\ 0 & \text{if } t \neq 0 \end{cases} \xRightarrow{\text{Generalised}} \delta(t-\tau) = \begin{cases} \infty & \text{if } t = \tau \\ 0 & \text{if } t \neq \tau \end{cases}$$

$\delta(t)$ is an infinately narrow, infinately tall idealised pulse with unit area

$$\boxed{\text{Property}: \int_{t_0-\varepsilon}^{t_0+\varepsilon} \delta(t-t_0)\, dt = 1}$$

Take note:

For DT, $\delta[n]$: Unit height at one discrete sample

For CT, $\delta(t)$: Unit area at one continuous time location

Sifting Formula for CT

Same concept as DT sifting formula, just that now its integrate and uses "unit area, $\delta(t)$" rather than "unit height, $\delta[n]$".

$$\boxed{x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau}$$

CT Impulse Response, h(t)

Same concept as discrete.

$$\delta(t) \longrightarrow \boxed{S} \longrightarrow h(t) \quad \text{OR} \quad h = S(\delta)$$

CT Convolution

Same concept as DT LTI.

$$\boxed{y(t) = x * h = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau}$$

Graphical CT Convolution

Same process as DT Convolution, just that "sum" become integrate

Reflect on $\tau = 0$ → Shift right by τ → Scale → Integrate

Properties of CT Convolution

Same as DT

1. $\delta * h = h$ [Identity]
2. $x * h = h * x$ [Commutative]
3. $h_{series} = h_1 * h_2$
4. $h_{parallel} = h_1 + h_2$
5. $S_1(S_2(x)) = S_2(S_1(x))$ [For single input single output CT LTI systems]

Step Response $s(t)$

The step response is the system output when input $= u(t)$

$$u(t) \longrightarrow \boxed{S} \longrightarrow s(t)$$

Remember / Realise that an integrator turns an impulse into a step.

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = u(t)$$

So similarly I can say integrator turns impulse response to step response.

$$\boxed{s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau}$$

OR

$$\boxed{h(t) = \frac{d}{dt} s(t)}$$

Reading System Properties from $h(t)$

Using the commutative convolution form

$$y(t) = \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

① Memoryless

Same concept as DT, just with "unit height, $\delta[n]$" $\longleftrightarrow$ "unit area, $\delta(t)$"

$$\boxed{h(t) = \alpha\, \delta(t)}$$

$\downarrow$ using $y(t) = x * h$

$$\boxed{y(t) = \alpha\, x(t)}$$

$\therefore$ Every memoryless LTI system is a gain system.

② Causality

Same concept as DT

$$\boxed{h(t) = 0 \quad , t < 0}$$

Rmb, a causal system cannot produce a response before receiving that sample at $t = 0$.

③ BIBO stability

In DT : Absolutely Summable

In CT : Absolutely Integrable

$$\boxed{\int_{-\infty}^{\infty} |h(\tau)| \, d\tau < 0}$$

DT vs CT LTI Summary Table

Conceptual job	Discrete Time (DT)	Continuous Time (CT)
Unit impulse	$\delta[n]$	$\delta(t)$
Key impulse idea	Height 1 at one sample	Unit area concentrated at one time
Sifting formula	$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$	$x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$
Dummy variable	k	τ
Impulse response	$\delta[n] \to h[n]$	$\delta(t) \to h(t)$
LTI response to shifted impulse	$\delta[n-k] \to h[n-k]$	$\delta(t-\tau) \to h(t-\tau)$
Scaled + shifted impulse response	$x[k]\delta[n-k] \to x[k]h[n-k]$	$x(\tau)\delta(t-\tau) \to x(\tau)h(t-\tau)$
Output / convolution	$y[n] = (x*h)[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$	$y(t) = (x*h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\,d\tau$
Convolution calculation	Flip → shift → multiply → sum	Flip → shift → multiply → integrate
Identity	$x*\delta = x$	$x*\delta = x$
Commutative	$x*h = h*x$	$x*h = h*x$
Series / cascade	$h_{\text{series}} = h_1 * h_2$	$h_{\text{series}} = h_1 * h_2$
Parallel	$h_{\text{parallel}} = h_1 + h_2$	$h_{\text{parallel}} = h_1 + h_2$
Memoryless	$h[n] = \alpha\delta[n]$	$h(t) = \alpha\delta(t)$
Causal	$h[n] = 0,\ n < 0$	$h(t) = 0,\ t < 0$
BIBO stable	$\sum_{k=-\infty}^{\infty} \lvert h[k]\rvert < \infty$	$\int_{-\infty}^{\infty} \lvert h(\tau)\rvert\,d\tau < \infty$
Name of stability condition	$h[n]$ is absolutely summable	$h(t)$ is absolutely integrable